



Samsung expanding its refurbished program

It could start selling newer refurbished phones soon. <https://dgit.in/oct22-09>



Twitter launches new features

Twitter launches new features to make video consumption on the platform better. <https://dgit.in/oct22-10>



ClearMR™ TIER

CMR RANGE

ClearMR 3000		$2500 \leq \text{CMR} < 3500$
ClearMR 4000		$3500 \leq \text{CMR} < 4500$
ClearMR 5000		$4500 \leq \text{CMR} < 5500$
ClearMR 6000		$5500 \leq \text{CMR} < 6500$
ClearMR 7000		$6500 \leq \text{CMR} < 7500$
ClearMR 8000		$7500 \leq \text{CMR} < 8500$
ClearMR 9000		$8500 \leq \text{CMR}$

pixels. If the pixels don't completely change state in a refresh cycle (16.7-milliseconds for 60Hz displays), the next frame will show some degree of lag that will manifest as motion blur in fast-moving sequences. As a prevailing industry standard, Blur for LCD and OLED displays are evaluated using Response Time which is often measured as the time taken for a pixel to go from peak black to peak white state. Manufacturers usually quote G2G response time and MPRT or motion picture response time as a spec to indicate motion clarity. G2G is the time it takes to switch from one shade of grey to another. This isn't a very good indicator since actual content often involves transitioning from different shades of grey or other colours, which may take longer. MPRT represents for how long a pixel will linger on the screen before it disappears. MPRT is dependent on frame rate and can be improved using motion enhancement techniques like Black Frame Insertion (BFI), overclocking, frame rate interpolation and more. Most LCD monitors today quote a response time of less than 1ms for the transition from one particular shade of grey to another, rendering the spec largely meaningless. Many of these LCD monitors can have a response time of more than a factor of 10 in practical scenarios. Besides, Manufacturers adopt techniques to enhance response time that introduces ugly artefacts in images. An OLED monitor with the same quoted response time will feel faster in practice.

So, yes, a metric that makes it easier for users to determine motion clarity is very much appreciated. This is easier said than done but that's what VESA plans on delivering with the new ClearMR certification.

HOW DOES VESA TEST A DISPLAY FOR CLEARMR CERTIFICATION?

VESA or its authorized labs use digital high-speed cameras to shoot pictures of test patterns as they switch from one frame to another. A colourimeter is used to verify the pattern luminance. Blur reduction methods like backlight strobing are disabled to maintain a level playing field. Some manufacturers increase the drive voltage and reduce response time for transitions and that may result in other overshoot artefacts. Vesa limits such image enhancement techniques while testing displays for CMR certification. The test is independent of the display technology being used and the final results are repeatable.

- All products are tested in SDR mode (HDR tests are under development).
- All products are tested at native resolution.
- All products are tested at their maximum frame rate.
- All products are tested at ambient room temperature.

CAN THE VESA CLEARMR CERTIFICATE BE TRUSTED FOR MOTION CLARITY?

It's still too early for us to comment

on how effective the VESA ClearMR certification will prove to be. VESA also has a DisplayHDR certification program which doesn't add much meaning for consumers but has been enthusiastically received by manufacturers. The certification doesn't properly define HDR. Screens with lower-tier certifications like the DisplayHDR400 and DisplayHDR600 for LCD panels do not produce impactful HDR. In the first set of VESA ClearMR certified monitors, we can see that LG's high-end Ultragear OLED monitors have the same ClearMR 7000 certification as IPS LCD monitors, which doesn't seem very convincing. We will have to wait and watch to know if this turns out to be more of a marketing logo or adds actual value for buyers.

TOP VESA CLEARMR CERTIFIED MONITORS TO BUY

LG was the first to come aboard and Samsung Displays will soon follow. As of now, a handful of LG monitors and HP Omen 25i have ClearMR certifications.



- LG 48" UltraGear UHD OLED Monitor - VESA ClearMR 7000
- LG 32" UltraGear QHD Nano IPS LCD Monitor - Vesa ClearMR 7000
- LG 27" UltraGear QHD Nano IPS LCD Monitor - Vesa ClearMR 7000
- LG 31.5" UltraGear QHD Nano IPS LCD Monitor - Vesa ClearMR 6000
- LG 32" UltraGear UHD 4K Nano IPS LCD Monitor - Vesa ClearMR 6000
- HP 32" Omen 25i IPS LCD Monitor - Vesa ClearMR 5000

You can find the official list at this link: <https://dgit.in/VESA>